Final Project

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2024-12-08

Introduction

Is there a significant difference in completion rates in full-time vs part-time first-time students in colleges in New York? The columns I will be using are;

i. OMAWDP6_FTFT (Percentage of full-time, first-time student receiving an award within 6 years of entry)
ii. OMAWDP6_PTFT (Percentage of part-time, first-time student receiving an award within 6 years of entry)
iii. STABBR (the state)
iv. INSTNM (the college's name)

This is a good topic because it looks at how being a full-time or part-time student affects graduation/completion rates. Focusing on colleges in New York makes it more specific and could help show how schools can better support different types of students. Since this is a test for differences in rates, I will be using a hypothesis test.

Preprocessing

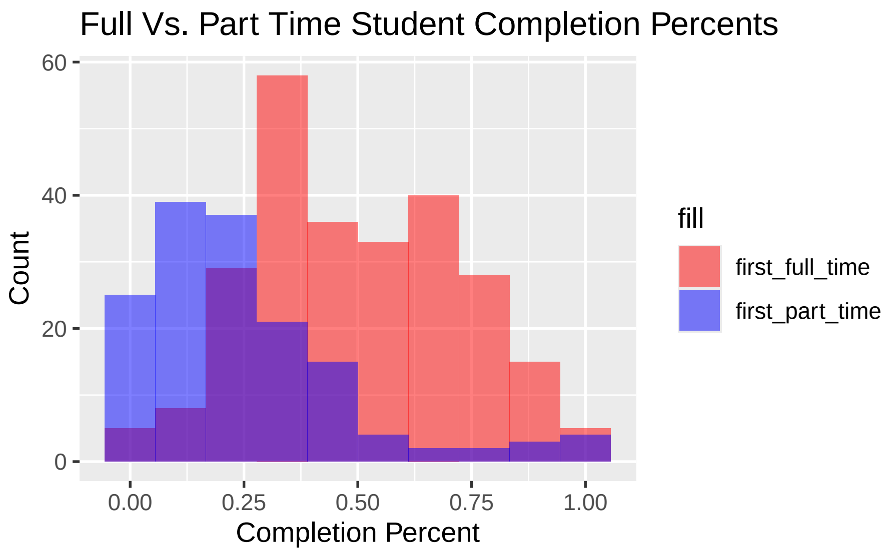
```r
ny_colleges <- college %>%
  select(OMAWDP6_PTFT, OMAWDP6_FTFT, STABBR, INSTNM) %>%
  filter(STABBR == "NY") %>%
  rename(first_full_time = OMAWDP6_FTFT,
         first_part_time = OMAWDP6_PTFT,
         state = STABBR,
         school_name = INSTNM)
```

I selected the OMAWDP6_PTFT, OMAWDP6_FTFT, STABBR, and INSTNM columns because these are the ones I need for my analysis. After selecting them, I filtered the data to include only colleges in New York (STABBR == 'NY'). Then, I renamed the columns to make them more understandable and easier to work with.

Visualization

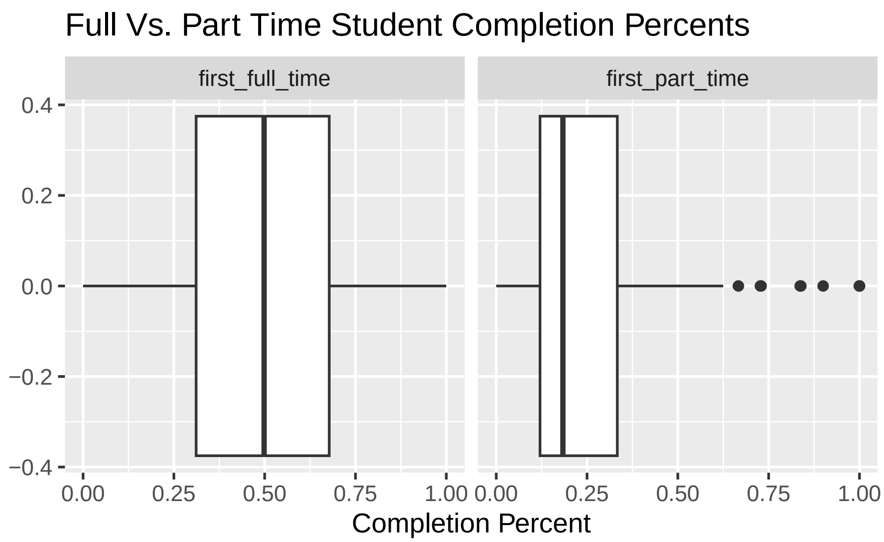
```r
ny_colleges %>%
  ggplot() +
  geom_histogram(aes(x = first_full_time, fill = "first_full_time"),
                 alpha = 0.5, bins = 10, na.rm = TRUE) +
```

```r
  geom_histogram(aes(x = first_part_time,fill = "first_part_time"),
                 alpha = 0.5, bins = 10, na.rm = TRUE) +
  scale_fill_manual(values = c("red", "blue")) +
  labs(title = "Full Vs. Part Time Student Completion Percents",
       x = "Completion Percent",
       y = "Count")
```



i. I picked this graph because the histograms show the completion rate for full-time and part-time first-time students, using color-coded overlays to reveal patterns.
ii. The histogram shows for the part-time students is unimodal and has a peak near 0.12. The count decreases quickly as the completion percentage increases. The histogram for the full-time students is bimodal and has a peak near 0.30. It decreases more steadily, meaning that full-time students have a broader range of completion rates compared to part-time students.

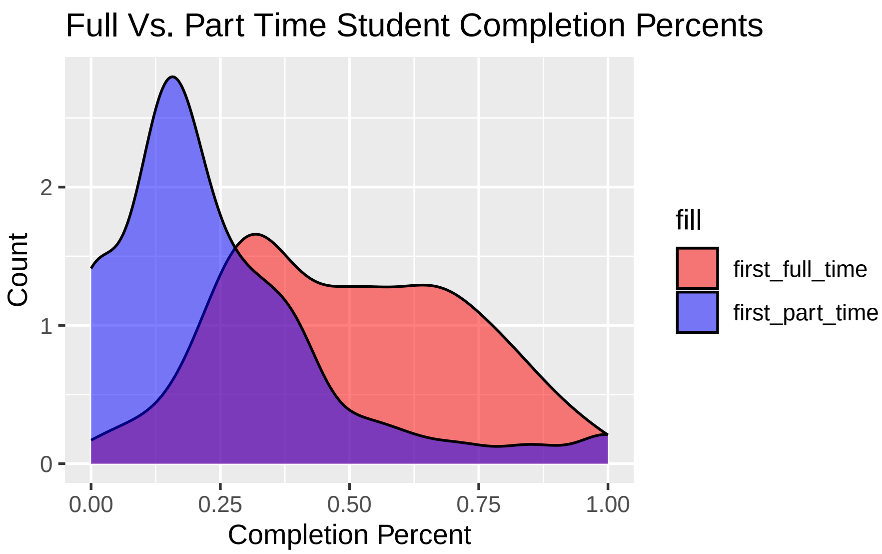
```r
ny_colleges %>%
  pivot_longer(cols = c(first_full_time, first_part_time),
               names_to = "student", values_to = "completion") %>%
  ggplot() +
  geom_boxplot(aes(x = completion), na.rm = TRUE) +
  facet_wrap(~student, scales = "free_x") +
  labs(title = "Full Vs. Part Time Student Completion Percents",
       x = "Completion Percent")
```



i. I wanted to see the distribution of values in a different way. The boxplots compare the spread and central tendency of completion rate for each group, showing differences in median, range, and outliers.

ii. The boxplot for full-time students has a wide range, spanning from 0 to approximately 2500, with a small number of outliers. Most outliers are close to the upper limit, but a few extend further, with one near 9000. The boxplot for part-time students is much narrower, concentrated near 0, and features a large number of outliers. Most of these outliers are close to the central range, with very few being far out. This means that full-time students have a broader range of completion rates compared to part-time students

```
ny_colleges %>%
  ggplot() +
  geom_density(aes(x = first_full_time, fill = "first_full_time"),
               alpha = 0.5, na.rm = TRUE) +
  geom_density(aes(x = first_part_time, fill = "first_part_time"),
               alpha = 0.5, na.rm = TRUE) +
  scale_fill_manual(values = c("red", "blue")) +
  labs(title = "Full Vs. Part Time Student Completion Percents",
       x = "Completion Percent",
       y = "Count")
```



i. I picked this graph because the density plot shows the distribution shape of the completion rate, smoothing the data to reveal trends like peaks or skewness, which helps visualize rate concentrations within specific ranges.

ii. The density plot smooths out the patterns seen in the histogram. The curve for part-time students peaks sharply near 0.12 and decreases quickly after, while the curve for full-time students has two peaks around 0.30 and 0.60. The graph shows a broader distribution with a slower decline, meaning that full-time students have a broader range of completion rates compared to part-time students.

Summary Statistics

```
ny_colleges %>%
  summarize(
    mean = mean(first_full_time, na.rm = TRUE),
    median = median(first_full_time, na.rm = TRUE),
    std_dev = sd(first_full_time, na.rm = TRUE),
    IQR = IQR(first_full_time, na.rm = TRUE),
    min_length = min(first_full_time, na.rm = TRUE),
    max_length = max(first_full_time, na.rm = TRUE)
  )
```

mean	median	std_dev	IQR	min_length	max_length
0.4986689	0.4984	0.2280905	0.3665	0	1

The full-time students show a mean completion rate of 0.4987, meaning on average, about 50% of full-time students complete their schools within six years. The median rate of 0.4984 suggests the data is symmetric around the average. With a standard deviation of 0.2281, the completion rates vary slightly, showing some differences across colleges. The IQR of 0.3665 and a range from 0 to 1 show the broader distribution of completion rates for full-time students.

```r
ny_colleges %>%
  summarize(
    mean = mean(first_part_time, na.rm = TRUE),
    median = median(first_part_time, na.rm = TRUE),
    std_dev = sd(first_part_time, na.rm = TRUE),
    IQR = IQR(first_part_time, na.rm = TRUE),
    min_length = min(first_part_time, na.rm = TRUE),
    max_length = max(first_part_time, na.rm = TRUE)
  )
```

mean	median	std_dev	IQR	min_length	max_length
0.2456039	0.1835	0.2216735	0.212775	0	1

The part-time students show a mean completion rate of 0.2456 so, only about 25% of part-time students complete their schools within six years. The median rate of 0.1835 is lower than the mean, meaning a right-skewed distribution where fewer students have high completion rates. A standard deviation of 0.2217 shows less variation compared to full-time students, meaning that rates are more concentrated. The IQR of 0.2128 and a range from 0 to 1 show that most part-time students fall within a narrower completion range.

Data Analysis

i. **Null Hypothesis (H0):** There is no difference in the mean completion rates between full-time and part-time first-time students in New York colleges.
ii. **Alternative Hypothesis (H1):** There is a difference in the mean completion rates between full-time and part-time first-time students in New York colleges.
iii. **Type of Test:** This is a two-sided test, as I am interested in seeing any difference, regardless of the direction.
iv. **Test Statistic:** The test statistic is the difference in mean completion rates between full-time and part-time students.

```r
ny_colleges_pivoted <- ny_colleges %>%
  pivot_longer(cols = c(first_full_time, first_part_time),
               names_to = "student_type",
               values_to = "completion_rate") %>%
  filter(!is.na(completion_rate))
```

```r
observed_stat <- ny_colleges_pivoted %>%
  specify(completion_rate ~ student_type) %>%
  calculate("diff in means", order = c("first_full_time", "first_part_time"))
observed_stat
```

stat
0.2530649

```r
null_distribution <- ny_colleges_pivoted %>%
  specify(completion_rate ~ student_type) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 10000, type = "permute") %>%
  calculate("diff in means", order = c("first_full_time", "first_part_time"))
```

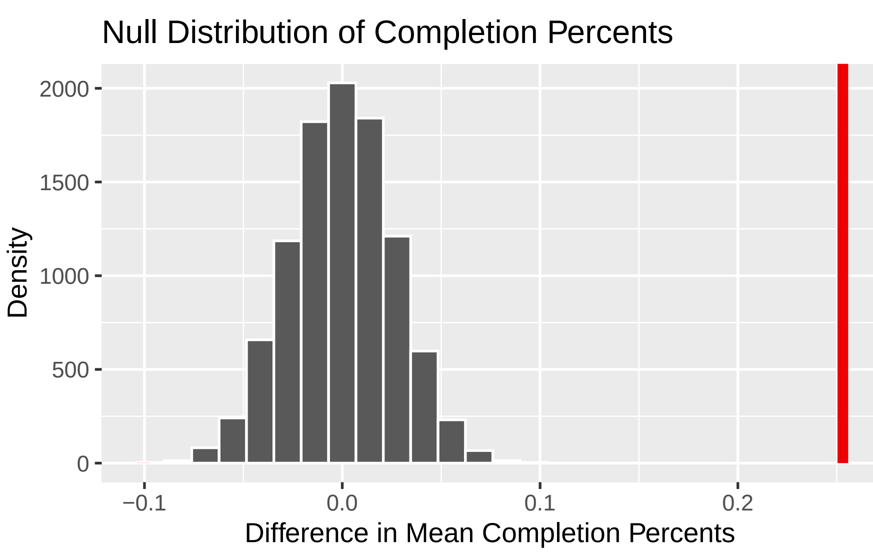
```r
p_value <- null_distribution %>%
  get_p_value(obs_stat = observed_stat, direction = "two-sided")
p_value
```

p_value
0

```r
if (p_value < 0.05) {
  conclusion <- "Reject the null hypothesis."
} else {
  conclusion <- "Fail to reject the null hypothesis."
}
conclusion
```

```
## [1] "Reject the null hypothesis."
```

```r
visualization <- visualize(null_distribution) +
  shade_p_value(obs = observed_stat, direction = "two.sided") +
  labs(title = "Null Distribution of Completion Percents",
       x = "Difference in Mean Completion Percents",
       y = "Density")
visualization
```



Based on the p-value of 0, I can reject the null hypothesis. This means that there is a statistically significant difference in the completion rates between full-time and part-time first-time students in New York colleges. Therefore, the data suggests that being a full-time or part-time student does impact the likelihood of completing a school within six years.

Conclusion

Based on the analyses performed, I can conclude that there is a significant difference in the completion rates between full-time and part-time first-time students in New York colleges. The visualizations, including histograms, boxplots, and density plots, clearly show that full-time students tend to have a broader range of completion rates, while part-time students have more concentrated and lower completion rates. The summary statistics further reinforce this observation, with full-time students having a higher mean and a wider spread in their completion rates. The hypothesis test results, with a p-value of 0, strongly support the rejection of the null hypothesis, indicating that being a full-time or part-time student does indeed affect graduation rates.

These findings directly answer the original question: there is a significant difference in completion rates between full-time and part-time first-time students. The visualizations, summary statistics, and hypothesis tests all support this conclusion. However, it's important to consider potential confounding factors. Variables such as the type of college, resources available, or students' socioeconomic backgrounds could also influence completion rates but are not included in the analysis. This study is valuable because understanding the factors influencing graduation rates can help colleges develop better support systems tailored to different student populations, ultimately improving educational outcomes.

Academic integrity statement

Online tools and guidance from SLACK was used to complete this assignment.